Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	
Project Name	Global Fuel Prices Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Access the global fuel price dataset (country-wise petrol, diesel and LPG prices in USD/litre, with date, region and currency fields) from CSV/Excel or public sources such as World Bank / GlobalPetrolPrices, and load it into the analysis tool.	High
2	Data Processing	Clean the data by removing duplicates, handling missing values, standardising country names and units, and converting currencies to USD. Add derived fields such as year, month, region and price change %.	High
3	Business Analysis	Calculate average, minimum and maximum fuel prices by country and region; year-on-year and month-on-month price change; ranking of most and least expensive countries; petrol vs diesel price gap; price volatility.	High
4	Business Insights & Analytical Outputs	Provide a dashboard with a world map of fuel prices, price trend line charts, top 10 / bottom 10 country bar charts, regional comparison charts and KPI cards for global average price and price change.	High
5	User Interaction & Analysis	Allow users to filter by country, region, fuel type and time period; compare multiple countries side by side; drill down from region to country to time trend; and hover for exact values.	Medium
6	Data Update & Reporting	Support refreshing the dashboard when a new dataset is loaded, and allow export of charts and summary reports as PDF/Excel for sharing with stakeholders.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboard should be simple and intuitive with clear labels, consistent colours and interactive filters, so non-technical users can understand fuel price trends without training.
NFR-2	**Security**	Only authorised users can access and edit the dashboard; data files are stored securely with role-based access. The dataset uses public data only and contains no personal information.
NFR-3	**Reliability**	The solution should give consistent and accurate results on every refresh, with data validation checks to catch errors, duplicates and missing values.
NFR-4	**Performance**	Dashboard pages and filters should load or update within 3-5 seconds, even when the full multi-year dataset for all countries is loaded.
NFR-5	**Availability**	The dashboard should be accessible whenever needed (target 99% uptime) through a web or cloud-based service, with backup copies of the dataset and reports.
NFR-6	**Scalability**	The solution should handle growth in data, such as more countries, years and fuel types, and more users, without redesigning the model or slowing down.